

TPC-390: Recursive Composition of a Frozen Count Slope

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Abstract

TPC-389 found a finite long-horizon transfer of a frozen count slope through $N = 1280$ on a fresh coordinate family. TPC-390 tests the next minimal composition: a fourth coordinate-disjoint family is calibrated at $N = 1024, 1280$ and evaluated at two holdout origins of length $N = 1536$. The 256-row, 32-cell panel separates a one-step parent forecast, a same-family local control, and a two-step application of the frozen parent slope. The recursive branch is deliberately judged against the same predeclared 3% finite cap. The one-step parent and local-control forecasts pass 30/32 and 32/32 cells, but recursive composition passes only 23/32, with maximum error 0.0490741652. The resulting finite pass/failure census is an obstruction audit, not an asymptotic law, an arithmetic estimate, or a twin-prime result.

1 Question and claim boundary

The previous releases transferred a finite logarithmic count-slope interface across disjoint origin families. The next natural question is whether that interface composes over another count layer. We choose, before reading any current response, the affine grid

$$a_j = 3000001 + 401j, \quad 0 \leq j < 41,$$

and retain indices 0, 10, 20, 30, 40. The first three origins are calibration origins at counts 1024 and 1280; the last two are holdouts at count 1536. The parent slopes are read from the hashed TPC-389 certificate and are not refit on this family.

The claim firewall is

$$\text{ARITHMETIC_ADVANCE=NO,} \quad \text{FIXED_POWER_CREDIT=0,} \quad \text{FULL_GATE_B=OPEN.}$$

The official Session evaluator files are absent in this checkout; the local Bridge-B is fail-closed repository evidence only.

2 Finite proxy

For $p \in (Q, 2Q]$ and $H = 66$, define

$$K_p(u, v) = p(p/Q)^2 \frac{H^2}{H^2 + (u - v)^2} \left(\mathbf{1}_{p|u-v} - \frac{1}{p-1} \right) \\ \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

The row geometry is the finite square energy $G(u) = \sum_p \sum_{v \in I} K_p(u, v)^2$. For each of four sign laws ℓ , the matrix is $M_\ell(u, v) = \sum_p s_\ell(p) K_p(u, v)$. We report local-diagonal normalization $M_\ell(u, v) / \sqrt{G(u)G(v)}$ and a pooled scalar normalization based only on calibration-origin geometry. The fixed band keeps block distance at most three; the full-relative band keeps all block pairs. We use $Q \in \{2048, 8192\}$ and block length 128.

Table 1: TPC-390 finite recursive-composition census.

quantity	result
parent one-step pass	30/32
local-control pass	32/32
recursive-composition pass	23/32
maximum parent one-step error	0.0363375462
maximum local-control error	0.0258044386
maximum recursive-composition error	0.0490741652
maximum composition-identity residual	3.33×10^{-16}
stable cells ($N = 1024, 1280, 1536$)	28/32, 25/32, 26/32
spectral failures / Schur failures	64/256, 0/256

3 Forecast interface

For a cell, let S_N be the mean band spectral diagnostic over its relevant origins. The parent exponent α_P is frozen from TPC-389. The local exponent uses only the current calibration pair:

$$\alpha_L = \frac{\log(S_{1280}/S_{1024})}{\log(1280/1024)}.$$

The one-step parent and local forecasts are

$$\widehat{S}_{1536}^P = S_{1280}(1536/1280)^{\alpha_P}, \quad \widehat{S}_{1536}^L = S_{1280}(1536/1280)^{\alpha_L}.$$

The recursive forecast is explicitly composed in two stages:

$$\widehat{S}_{1280}^{(1)} = S_{1024}(1280/1024)^{\alpha_P}, \quad \widehat{S}_{1536}^{(2)} = \widehat{S}_{1280}^{(1)}(1536/1280)^{\alpha_P}.$$

The direct expression $S_{1024}(1536/1024)^{\alpha_P}$ is retained as a composition identity control. Each holdout ratio is the observed S_{1536} divided by its declared forecast, minus one; a finite pass requires absolute error at most 0.03.

4 Certification and finite result

The canonical certificate contains 256 rows and 32 cells. The producer sums the prime shell in ascending order. An independent implementation rebuilds the same finite matrices in descending shell order, recomputes the row metrics, and reconstructs every forecast cell from the rows. A rational 13-point $Q = 8$ anchor at [3000001, 3000014] checks positive geometry and symmetry for all four laws. A 25-mutation stress suite rejects altered provenance, roles, rows, summaries, and firewall values.

The exact numerical census below is copied from the sealed canonical certificate; all decimal values are finite diagnostics, not claimed limits.

The composition identity itself has residual at most $3.3306690738754696 \times 10^{-16}$, so the obstruction is not an associativity artifact. The largest recursive failure is fixed-three-block, pooled, alternating-index at $Q = 2048$. Recursive failures also occur in the fixed-three-block pooled all-plus, mod-4, and half-split cells, and for alternating-index at $Q = 2048$ under local-diagonal and full-relative normalizations. The one-step parent failures are concentrated in the fixed-three-block pooled and full-relative pooled alternating-index $Q = 2048$ cells.

The interpretation is intentionally conditional on this finite panel. A recursive cap failure identifies a horizon-dependent obstruction for this declared proxy and normalization; it does not refute a different source-valid theorem. Conversely, a finite pass would not establish count or origin uniformity. In either case the inherited spectral diagnostic and the open arithmetic gates remain separate.

5 Conclusion and next clue

TPC-390's reusable object is a frozen parent slope with an explicit two-stage composition audit on a fresh family. The strongest positive result is the local 32/32 control and the numerically exact composition identity; the strongest obstruction is the 23/32 recursive pass census and its 4.91% maximum error. The next clue is

ROUND2_CLUE=LOCALIZE_C1_RECURSIVE_HORIZON_OBSTRUCTION.

No arithmetic power credit is assigned, and Route-A/Route-B reassembly and the twin-prime endpoint remain open.

Reproduction

All source, certificate, proof, experiment, and Bridge-B files are stored under `papers/tpc-390-c1-recursive-slope-composition/`. Run the ordinary and optimized producer, independent checker, mutation stress, and local Bridge-B commands listed in the project README. The release uses `paper/main.pdf` and `paper/paper.pdf` as byte-identical copies.